

Homework Assignment 8

1. An electrical rocket vehicle has the following characteristics:
 $(M_L/M_0) = 0.7$
 $I_{sp} = 3500 \text{ s}$, $t_b = 12 \text{ weeks}$, $\alpha = 5 \text{ kg/kW}$, $\eta = 0.5$.
Calculate $\Delta u_{b,imp}$. {Ans.: 6.214 km/s}
2. Holding the “burn” time, specific mass and efficiency at the values specified, and the velocity rise at the value calculated, find the optimum specific impulse for $(M_L/M_0)_{max}$.
3. An electrical rocket vehicle has the following characteristics:
 $\dot{\phi} = 2.5 \text{ kW}$, $\alpha = 5 \text{ kg/kW}$, $\eta = 0.75$, $I_{sp} = 3000 \text{ s}$
 $M_P = 51 \text{ kg}$
 $M_L = 34 \text{ kg}$
Find the “burn” time, the thrust and $\Delta u_{b,imp}$. {Ans.: 136.33 days, 127.4 mN, 21.79 km/s}